# Quantum Attention Is a Kernel, Not a Speedup: Training-Cost Bounds and Linear-Time Recall for Transformers

Chandra Sekhar Singuru
*Independent Researcher*
Long Beach, CA, USA
sekharc669@gmail.com
ORCID: 0009-0004-0219-1960

***Abstract*—Quantum self-attention computes Transformer attention scores as overlaps of quantum states. We ask whether this is worth doing on quantum hardware, and what classical Transformers should borrow from it. We measure two scaling laws of the fidelity attention score: its mean decays as $2^{-n}$ and its gradient variance as approximately $4^{-n}$ in the number of qubits $n$ (fitted rate $3.97$, 95% interval $3.85$–$4.06$). Combining them with Bernoulli shot noise and parameter-shift gradients, we prove a lower bound on the shots per training step and a closed-form condition for any quantum advantage over exact classical simulation. The minimum training time inside any advantage regime scales with the inverse square of the hardware shot rate: at least 64 years per step on one sequence at $10^4$ shots/s and 1.5 days at $10^6$, and a sweep over 72 hardware scenarios confirms that a $10^4$-step training run is out of reach in every case. On the classical side, we give an exact $3^n$-dimensional feature map for the cosine-product kernel that underlies fidelity attention and evaluate the resulting linear-time layer, CPLA, on multi-query associative recall in two-layer Transformers. At an identical 81-dimensional state, CPLA retains 86% recall at 16 key-value pairs where standard linear attention falls to 12.5%, and it far exceeds Performer; it trails the second-order Taylor kernel of Based by 4–13 points at moderate load, and no fixed-state method survives 32 pairs. Quantum attention thus contributes a useful multiplicative kernel, not a computational speedup.**

***Index Terms*—Transformers, attention mechanism, linear attention, associative recall, quantum machine learning, quantum kernels, barren plateaus, computational cost**

## I. Introduction

Attention is the most expensive part of a Transformer [1]. Its pairwise scores cost time quadratic in the sequence length $N$, which has motivated a large body of work on efficient attention: kernel views of attention [2], linear attention [3], random-feature approximations [4] and re-weighted linear kernels [5]. A recurring finding is that efficient models struggle with *associative recall*, retrieving a value seen earlier in the context given its key [6], [7].

Quantum self-attention has been proposed as another route [8]–[13]. Tokens are encoded as quantum states, and attention scores are computed from state overlaps on a quantum processor. Our earlier work [14] showed that, for angle-encoded states, the fidelity score equals a classical cosine-product (CP) kernel, and that any fidelity attention admits an exact linear-time classical evaluation. That result leaves two questions that matter to practitioners building Transformers.

1) **Is training quantum attention on hardware ever worth its cost?** Existing analyses estimate shot counts for evaluating quantum kernels [15], [16], but not the cost of *training* an attention layer, where every token pair needs gradient estimates.
2) **What, if anything, should classical Transformers borrow?** If the useful part of quantum attention is a classical kernel, does that kernel help on the recall tasks where efficient attention is known to fail?

Our contributions are:

- **Two measured scaling laws** for the quantum fidelity score: its mean decays as $2^{-n}$ and its gradient variance as $\approx 4^{-n}$ in the number of qubits $n$ (Section IV).
- **A training-cost theorem.** Accounting for Bernoulli shot noise, parameter-shift gradients and the most quantum-favourable aggregation of gradients, we derive a lower bound on the shots per training step and a closed-form condition for any quantum advantage over exact classical simulation (Section V).
- **A quantitative verdict.** Any regime in which quantum hardware is cheaper than exact classical simulation needs at least 1.5 days per training step on a single sequence at $10^6$ shots/s, and at least 64 years at $10^4$ shots/s; a numerical sweep over 72 hardware scenarios agrees.
- **An exact $3^n$-dimensional feature map** for the CP kernel, smaller than the $4^n$ of [14], and an evaluation of the resulting linear-time layer (CPLA) on multi-query associative recall against Based [7], linear attention and Performer (Section VI).

## II. Related Work

### A. Efficient Attention and Recall

Viewing attention as a kernel smoother [2] separates the similarity function from the rest of the layer. Linear attention [3] uses a positive feature map so that the sum over keys can be reassociated, giving cost linear in $N$; Performer [4] approximates the softmax kernel with random features, and

cosFormer [5] modulates scores with a cosine of token distance. Zoology [6] introduced multi-query associative recall (MQAR) and showed that efficient architectures lag attention on it. Based [7] combines a small sliding window with linear attention built on a second-order Taylor approximation of the exponential, trading recurrent-state size for recall.

### B. Quantum Self-Attention

QSANN [8] scores queries and keys from single Pauli-$Z$ measurements; QKSAN [9] uses quantum kernel overlaps; QMSAN [10] uses swap tests between mixed states; Pecilli and Rosati [11] drop the softmax to obtain gate complexity linear in sequence length; and Xu *et al.* [12] build higher-order interactions from expectation read-outs. A recent survey [13] notes that pairwise quantum attention keeps the quadratic dependence on $N$. Our prior work [14] reduced fidelity attention exactly to classical computation. None of these works analyses the end-to-end cost of training.

### C. Cost and Trainability of Quantum Kernels

Fidelity kernels [17] can concentrate exponentially, so that distinguishing inputs needs exponentially many shots [16]. Slattery *et al.* [18] found that well-tuned fidelity kernels are approximated by classical kernels, and Bowles *et al.* [19] found no systematic advantage of quantum models in benchmarks. Miroszewski *et al.* [15] estimated the shots needed to *evaluate* quantum kernel matrices and compared them with classical simulation, concluding that quantum utility may appear above about 40 qubits. Barren plateaus [20], particularly for global observables [21], make gradients vanish exponentially, and their provable absence often coincides with classical simulability [22]. We add the attention-specific, training-time analysis: quadratic pairs, gradient estimation, and comparison against the linear-time classical reduction.

## III. Fidelity Attention and Its Classical Reduction

Each token embedding $x$ is mapped to $n$ angles $\phi(x) = \pi \tanh(Ax)$ and encoded as $S(\phi)\,|0\rangle^{\otimes n} = \bigotimes_k (\cos\frac{\phi_k}{2}\,|0\rangle + \sin\frac{\phi_k}{2}\,|1\rangle)$. Separate hardware-efficient ansatzes $W_Q$, $W_K$ with $L$ layers of $R_Y$ rotations and CNOT rings give query and key states, and the score is the fidelity

$$s_{ij} = |\langle\psi_K(x_j)|\psi_Q(x_i)\rangle|^2\,, \tag{1}$$

estimated on hardware with a compute–uncompute circuit whose all-zeros outcome probability equals $s_{ij}$ [14]. Each score circuit has $P = 2n + 2nL = 2n(L+1)$ rotation angles, and gradients follow the parameter-shift rule $\partial s/\partial a = \frac{1}{2}[s(a+\frac{\pi}{2}) - s(a-\frac{\pi}{2})]$ [23], so a training step evaluates $1+2P$ circuits per token pair.

Without an ansatz the score is the CP kernel $s_{ij} = \prod_k \cos^2\big((\phi_{ik} - \phi_{jk})/2\big)$, and with any ansatz $s_{ij}$ is an inner product of explicit $4^n$-dimensional features [14]. We sharpen the feature dimension for the CP kernel.

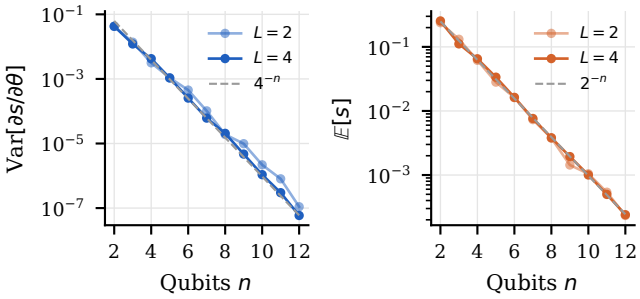



Fig. 1. Gradient variance (left) and mean score (right) of the fidelity attention score at uniformly random parameters, 400 samples per point, for $L = 2$ and $L = 4$ ansatz layers. Dashed lines show $4^{-n}$ and $2^{-n}$.

**Proposition 1** (Exact $3^n$ feature map)**.** *Let* $f(\phi) = \bigotimes_{k=1}^{n} \frac{1}{\sqrt{2}}\,[1, \cos\phi_k, \sin\phi_k] \in \mathbb{R}^{3^n}$. *Then* $\prod_k \cos^2\big((a_k - b_k)/2\big) = \langle f(a), f(b)\rangle$.

*Proof.* For one feature, $\cos^2\frac{a-b}{2} = \frac{1}{2}\big(1+\cos(a-b)\big) = \frac{1}{2}\big(1+\cos a\cos b+\sin a\sin b\big)$, the inner product of $\frac{1}{\sqrt{2}}[1, \cos a, \sin a]$ and $\frac{1}{\sqrt{2}}[1, \cos b, \sin b]$. Inner products of tensor products factorize. □

The map reduces the linear-time state from $4^n$ to $3^n$ per head (81 instead of 256 for $n = 4$), and allows separate query and key angle projections while preserving exact linear-time evaluation. We call the resulting layer cosine-product linear attention (CPLA).

## IV. Two Scaling Laws of the Fidelity Score

We measured, for $n = 2$ to 12 and $L \in \{2, 4\}$, the mean score $p(n) = \mathbb{E}[s]$ and the gradient variance $V(n) = \mathrm{Var}[\partial s/\partial\theta_{Q,1}]$ over uniformly random inputs and parameters, using an exact statevector simulator validated against the Qiskit Estimator [24] to within $10^{-15}$. Fitting log-linear models for $n \geq 4$ gives (95% bootstrap intervals in parentheses):

- $p(n) \propto 1.99^{-n}$ ($L = 2$) and $2.01^{-n}$ ($L = 4$), matching the Haar value $\mathbb{E}|\langle\psi|\varphi\rangle|^2 = 2^{-n}$;
- $V(n) \propto 3.53^{-n}$ (3.27–3.83) for $L = 2$ and $3.97^{-n}$ (3.85–4.06) for $L = 4$.

The deeper circuit reaches the $4^{-n}$ rate expected of a global observable on an approximate 2-design [20], [21]. Fig. 1 shows both laws. They are the inputs to the cost analysis, and we state their extrapolation beyond 12 qubits as an explicit assumption.

**Assumption 1.** For the register sizes considered, $p(n) = c_p\, r_p^{\,n}$ and $V(n) = a\, r_V^{\,n}$ with the fitted constants, and a typical per-pair gradient has magnitude $\sqrt{V(n)}$.

## V. The Cost of Training Fidelity Attention

We compare one training step (forward and backward) on one sequence of $N$ tokens. Let $R_s$ be the hardware shot rate, $F$ the classical floating-point rate, $d$ the value width and $g = n(L+1)$ the gates per state.

**Lemma 1** (Shot noise)**.** *A score estimated from $S$ shots has variance $p(1-p)/S$, and the parameter-shift estimate $\hat{g} = \frac{1}{2}(\hat{s}_+ - \hat{s}_-)$ has variance $p(1-p)/(2S)$.*

*Proof.* Each shot is a Bernoulli trial with success probability $s \approx p$. The two shifted estimates are independent, so $\operatorname{Var}\hat{g} = \frac{1}{4} \cdot 2\,p(1-p)/S$. □

**Theorem 1** (Quantum shots per step)**.** *Under Assumption 1, resolving the gradient of a shared parameter at signal-to-noise ratio at least one requires, per training step,*

$$Q \;\geq\; (1+2P)\,\max\Big(N^2,\; \frac{p(1-p)}{2V}\Big) \tag{2}$$

*shots, even if per-pair gradients add perfectly coherently.*

*Proof.* Each of the $N^2$ pairs needs at least one shot on each of its $1+2P$ circuits, giving the first term. The aggregated gradient is $G = \sum_{ij} w_{ij}\hat{g}_{ij}$, with noise variance $\sum w_{ij}^2\, p(1-p)/(2S)$ by Lemma 1. Its signal satisfies $|\mathbb{E}G|^2 \leq (\sum |w_{ij}|)^2 V \leq N^2 \sum w_{ij}^2 V$ by Cauchy–Schwarz, with equality for perfectly coherent pairs. Signal-to-noise at least one therefore needs $S \geq p(1-p)/(2N^2V)$ shots per circuit, and the total is $N^2(1+2P)S$. □

If per-pair gradients are incoherent, the signal grows only as $N$ and the second term in (2) is multiplied by $N^2$.

**Theorem 2** (Exact classical cost)**.** *Exact forward and backward evaluation of fidelity attention takes at most*

$$T_c = \frac{6}{F}\min\big(2Ng2^n + 2N^2 2^n,\; 2Ng2^n + 4N\,4^n d\big) \tag{3}$$

*seconds, with memory $2^{n+5}$ bytes for the first (pairwise) algorithm and $8 \cdot 4^n d$ bytes for the second (linear-time) algorithm.*

*Proof.* Preparing a state costs $g$ gate applications on $2^n$ amplitudes. The pairwise algorithm computes $N^2$ overlaps of length $2^n$, streaming states so that two are held at once. The linear-time algorithm uses the explicit $4^n$ features of [14] with prefix sums. Reverse-mode differentiation costs at most twice the forward pass, giving the factor $6 = 2\times 3$ for multiply-adds over forward and backward. □

**Corollary 1** (Necessary condition for advantage)**.** *If quantum hardware is faster than the pairwise classical algorithm, then $2^n(1+g/N) > (1+2P)\,F/(12R_s)$.*

*Proof.* By Theorem 1, $T_q \geq (1+2P)N^2/R_s$; by Theorem 2, $T_c \leq 12(N^2+Ng)2^n/F$. Require $T_q < T_c$. □

**Corollary 2** (Minimum time per step)**.** *If $p = 2^{-n}$ and $V = a\,4^{-n}$, then whenever quantum hardware beats exact classical simulation with sufficient classical memory,*

$$T_q \;\geq\; \frac{(1+2P)^2\,F}{24\,a\,R_s^2}\cdot\frac{N}{N+g}. \tag{4}$$

*Proof.* By Theorem 1, $T_q \geq (1+2P)\,p(1-p)/(2VR_s) \approx (1+2P)\,2^n/(2aR_s)$. Substituting the lower bound on $2^n$ from Corollary 1 gives (4). □

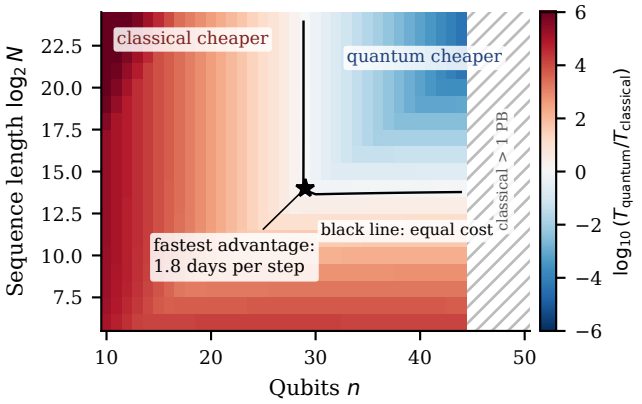



Fig. 2. Ratio of quantum to exact classical training time per step for $L = 4$, $R_s = 10^6$ shots/s, $F = 10^{13}$ flop/s and 1 PB classical memory. Quantum is cheaper only in the upper right, and the fastest such configuration (star) still needs 1.8 days per step on one sequence.

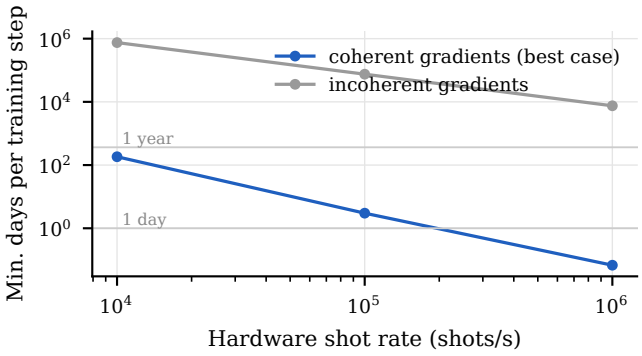



Fig. 3. Minimum time per training step over all configurations where quantum hardware beats exact classical simulation, versus hardware shot rate (sweep over $L$, classical flop rate $10^{13}$–$10^{17}$ and memory $10^{12}$–$10^{15}$ bytes).

The bound grows as $1/R_s^2$: hardware shot rate enters quadratically, once through the per-pair floor and once through the gradient-shot requirement. With the measured constants for $L = 4$ ($a = 1.07$) and $F = 10^{13}$ (one workstation), the smallest qualifying register and the resulting bound are $n = 36$ and $T_q \geq 64$ years per step at $R_s = 10^4$ shots/s; $n = 33$ and 198 days at $10^5$; and $n = 29$ and 1.5 days at $10^6$.

*Numerical sweep.* To avoid relying on the idealized rates, we evaluated (2) and (3) with the fitted constants for every $n \leq 60$ and $N = 2^6$ to $2^{24}$, across 72 scenarios: $L \in \{2, 4\}$, $R_s \in \{10^4, 10^5, 10^6\}$, $F \in \{10^{13}, 10^{15}, 10^{17}\}$, memory $\in \{1\,\text{TB}, 1\,\text{PB}\}$ and coherent or incoherent gradients. Fig. 2 maps the ratio for one scenario and Fig. 3 summarizes all of them. A quantum advantage region always exists at large $n$ and $N$, but the fastest training step inside it takes at least 0.5 years at $10^4$ shots/s, 3.0 days at $10^5$ and 1.6 hours at $10^6$ with coherent gradients; with incoherent gradients the minima are 2,060, 206 and 20.6 years. A modest run of $10^4$ steps is therefore out of reach in every scenario, while the corresponding classical simulation remains tractable or is the

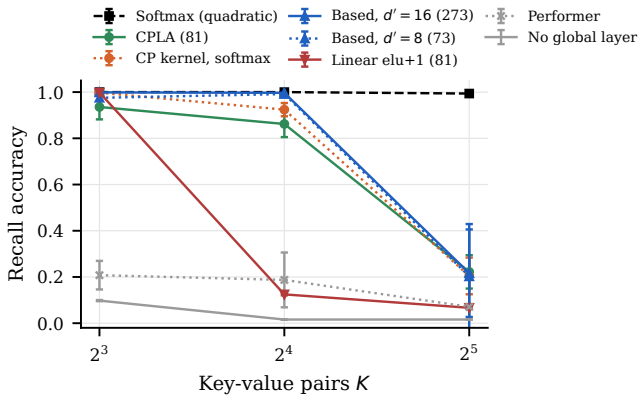



Fig. 4. Associative recall accuracy versus number of key–value pairs (mean ± std., 3 seeds). Numbers in parentheses are recurrent-state dimensions per head. At the same state size, CPLA (green) retains recall at $K = 16$ where linear elu+1 attention (red) collapses.

cheaper option.

## VI. What Transformers Can Borrow: Associative Recall

The cost analysis implies that the value of quantum attention, if any, lies in its kernel. We therefore test the classical CP kernel where efficient attention is known to be weak: multi-query associative recall (MQAR) [6].

### A. Setup

Each sequence lists $K$ key–value pairs followed by $\min(K, 16)$ queries, each a previously seen key; the model must output the paired value. Keys and values come from disjoint vocabularies of 64 tokens each, so chance accuracy is $1/64$. Following Based [7], every model is a two-layer, four-head, pre-norm causal Transformer ($d = 64$) whose first layer is sliding-window softmax attention over the last 4 tokens and whose second layer is the attention variant under test. A control with sliding windows in both layers measures how much recall the global layer provides. The second layer is one of:

- **softmax**: full quadratic attention, the reference;
- **CPLA**: linear-time CP kernel with separate 4-angle query and key projections per head, state $3^4 = 81$ (Proposition 1);
- **CP softmax**: the same kernel inside a softmax, quadratic;
- **Based** [7]: second-order Taylor features of $\exp(q{\cdot}k/\sqrt{d'})$, state $1 + d' + d'^2 = 73$ ($d' = 8$) or 273 ($d' = 16$);
- **linear elu+1** [3] with state 81, matched to CPLA;
- **Performer** [4] with 64 positive random features.

All models train for 2,000 steps of AdamW (batch 64, learning rate $10^{-3}$ with one-cycle schedule) on freshly sampled sequences and are evaluated on 2,048 held-out sequences, over three seeds for $K \in \{8, 16, 32\}$. Causality of every variant and exactness of the linear-time forms against their pairwise counterparts were verified numerically.

TABLE I
Multi-Query Associative Recall Accuracy (Mean ± Std., 3 Seeds)

| Second layer | State | $K = 8$ | $K = 16$ | $K = 32$ |
|---|---|---|---|---|
| Softmax (quadratic) | $O(N)$ | 1.000±.000 | 1.000±.000 | .994±.002 |
| CP softmax (quadratic) | $O(N)$ | .997±.001 | .924±.029 | .204±.079 |
| Based, $d'$=16 | 273 | .999±.001 | .997±.002 | .216±.189 |
| Based, $d'$=8 | 73 | .975±.032 | .991±.004 | .203±.226 |
| CPLA (ours) | 81 | .936±.054 | .862±.057 | .222±.072 |
| Linear elu+1 | 81 | .998±.001 | .125±.004 | .067±.002 |
| Performer | 64 | .208±.062 | .187±.119 | .071±.002 |
| No global layer | – | .098±.002 | .016±.001 | .016±.000 |

### B. Results

Table I and Fig. 4 give the results. The control without a global layer stays near chance, so recall is performed by the second layer.

*Kernel shape matters at fixed state size.* CPLA and linear elu+1 attention carry the same 81-dimensional state per head, yet at $K = 16$ CPLA recalls 86.2% of values and elu+1 only 12.5%, a paired difference of $0.74 \pm 0.06$ over seeds. CPLA also exceeds Performer by 0.68 at $K = 16$. The CP kernel multiplies agreement across angle features, so a query matches a key only if all features align, whereas elu+1 features combine additively. Based's second-order Taylor kernel shares this multiplicative character, and both retain recall where additive features fail.

*Based remains stronger.* At moderate load Based outperforms CPLA by 4.0 points at $K = 8$ and 12.9 at $K = 16$ with a smaller 73-dimensional state; at $K = 32$ the two are indistinguishable. CPLA is also within 6.2 points of its own softmax version, so the linear-time form costs little relative to the kernel itself.

*No fixed state survives high load.* At $K = 32$ every linear-time variant falls to 7–22% while softmax attention keeps 99.4%, consistent with the recall–state-size trade-off of [6], [7]. At $K = 64$ even softmax attention reached only 40.7% within the 2,000-step budget, so we do not report that setting.

## VII. Discussion and Limitations

*For AI practitioners.* Quantum fidelity attention contributes a kernel, not a computational resource. Its CP form is exactly computable in linear time with $3^n$ features per head, and its usefulness should be judged against efficient-attention baselines at matched state size, as we did here.

*For quantum machine learning.* The cost theorem isolates three levers: the per-pair floor $N^2(1 + 2P)$, the gradient-shot requirement $p/(2V)$, and the shot rate. Escaping the conclusion would require at least one of: gradient variance decaying more slowly than the score mean squared, which in practice means local rather than global observables [21]; a non-pairwise readout that avoids $N^2$ circuits [11]; or shot rates far above $10^6$ per second.

*Limitations.* The scaling laws are measured up to 12 qubits and extrapolated (Assumption 1). The cost model counts shots and floating-point operations and ignores error correction,

compilation, communication and classical-quantum latency, all of which would add to the quantum side. The recall experiments use small two-layer models and synthetic data. The recall gap to softmax attention at $K = 32$ and the failure of all models at $K = 64$ reflect a fixed 2,000-step budget; longer training or larger states may change the ranking.

## VIII. Conclusion

Quantum fidelity attention is best understood as a kernel, not a speedup. Its score mean and gradient variance decay as $2^{-n}$ and $4^{-n}$, and with these laws we proved that any regime where hardware beats exact classical simulation demands a training time that grows with the inverse square of the shot rate, from 1.5 days per step at $10^6$ shots/s to 64 years at $10^4$. What Transformers can borrow is the cosine-product kernel itself: computed exactly in linear time with $3^n$ features, it retains associative recall at a fixed state size where additive linear attention collapses, although Based's Taylor kernel remains stronger. Future work should combine CP and Taylor features, test the kernel in multi-layer language models, and look for quantum readouts that are local and non-pairwise, the only routes our bounds leave open.

## Code and Data Availability

Code for the scaling measurements, cost analysis and recall experiments is provided as supplementary material.

## Disclosure of AI Assistance

Generative AI tools were used to assist with code development, experiment design and drafting of this manuscript. The author reviewed, verified and takes full responsibility for all content.

## References


[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[2] Y.-H. H. Tsai, S. Bai, M. Yamada, L.-P. Morency, and R. Salakhutdinov, "Transformer dissection: An unified understanding for Transformer's attention via the lens of kernel," in *Proc. EMNLP-IJCNLP*, 2019, pp. 4344–4353.

[3] A. Katharopoulos, A. Vyas, N. Pappas, and F. Fleuret, "Transformers are RNNs: Fast autoregressive transformers with linear attention," in *Proc. 37th Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 119, 2020, pp. 5156–5165.

[4] K. Choromanski, V. Likhosherstov, D. Dohan *et al.*, "Rethinking attention with Performers," in *Int. Conf. Learning Representations (ICLR)*, 2021.

[5] Z. Qin, W. Sun, H. Deng *et al.*, "cosFormer: Rethinking softmax in attention," in *Int. Conf. Learning Representations (ICLR)*, 2022.

[6] S. Arora, S. Eyuboglu, A. Timalsina, I. Johnson, M. Poli, J. Zou, A. Rudra, and C. Ré, "Zoology: Measuring and improving recall in efficient language models," in *Int. Conf. Learning Representations (ICLR)*, 2024.

[7] S. Arora, S. Eyuboglu, M. Zhang, A. Timalsina, S. Alberti, J. Zou, A. Rudra, and C. Ré, "Simple linear attention language models balance the recall-throughput tradeoff," in *Proc. 41st Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 235, 2024, pp. 1763–1840.

[8] G. Li, X. Zhao, and X. Wang, "Quantum self-attention neural networks for text classification," *Science China Information Sciences*, vol. 67, no. 4, p. 142501, 2024.

[9] R.-X. Zhao, J. Shi, and X. Li, "QKSAN: A quantum kernel self-attention network," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 12, pp. 10 184–10 195, 2024.

[10] F. Chen, Q. Zhao, L. Feng, C. Chen, Y. Lin, and J. Lin, "Quantum mixed-state self-attention network," arXiv:2403.02871, 2024.

[11] A. Pecilli and M. Rosati, "Quantum attention by overlap interference: Predicting sequences from classical and many-body quantum data," arXiv:2602.06699, 2026.

[12] J. Xu, C. Li, D. Zeng, J. Paisley, and Q. Zhao, "Higher-order token interactions via quantum attention," arXiv:2606.11673, 2026.

[13] H. Zhang, Q. Zhao, M. Zhou, L. Feng, D. Niyato, S. Zheng, and L. Chen, "A survey of quantum transformers: Architectures, challenges and outlooks," arXiv:2504.03192, 2025.

[14] C. S. Singuru, "From quantum fidelity to linear-time attention: An exact classical reduction of quantum self-attention," https://github.com/Chandra953/Quantum-Fidelity-Attention, 2026.

[15] A. Miroszewski, M. Fellous Asiani, J. Mielczarek, B. Le Saux, and J. Nalepa, "In search of quantum advantage: Estimating the number of shots in quantum kernel methods," arXiv:2407.15776, 2024.

[16] S. Thanasilp, S. Wang, M. Cerezo, and Z. Holmes, "Exponential concentration in quantum kernel methods," *Nature Communications*, vol. 15, p. 5200, 2024.

[17] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, and J. M. Gambetta, "Supervised learning with quantum-enhanced feature spaces," *Nature*, vol. 567, no. 7747, pp. 209–212, 2019.

[18] L. Slattery, R. Shaydulin, S. Chakrabarti, M. Pistoia, S. Khairy, and S. M. Wild, "Numerical evidence against advantage with quantum fidelity kernels on classical data," *Physical Review A*, vol. 107, p. 062417, 2023.

[19] J. Bowles, S. Ahmed, and M. Schuld, "Better than classical? The subtle art of benchmarking quantum machine learning models," arXiv:2403.07059, 2024.

[20] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, "Barren plateaus in quantum neural network training landscapes," *Nature Communications*, vol. 9, p. 4812, 2018.

[21] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, and P. J. Coles, "Cost function dependent barren plateaus in shallow parametrized quantum circuits," *Nature Communications*, vol. 12, p. 1791, 2021.

[22] M. Cerezo, M. Larocca, D. García-Martín, N. L. Diaz, P. Braccia, E. Fontana, M. S. Rudolph, P. Bermejo, A. Ijaz, S. Thanasilp, E. R. Anschuetz, and Z. Holmes, "Does provable absence of barren plateaus imply classical simulability?" *Nature Communications*, vol. 16, p. 7907, 2025.

[23] M. Schuld, V. Bergholm, C. Gogolin, J. Izaac, and N. Killoran, "Evaluating analytic gradients on quantum hardware," *Physical Review A*, vol. 99, p. 032331, 2019.

[24] A. Javadi-Abhari, M. Treinish, K. Krsulich, C. J. Wood *et al.*, "Quantum computing with Qiskit," arXiv:2405.08810, 2024.